MineMirror AI: Uniqueness, India Position and Benefits

Prepared 31 October 2026 from the MineMirror AI website and public sources (DGMS, Lok Sabha records, company press releases).

1. Executive summary

MineMirror AI presents itself as a real-time, physics-informed digital twin platform for mining. It aims to predict geotechnical hazards, track equipment health and monitor worker exposure to dust, vibration and air quality. Its founder and CEO is Nitish Kumar Roy, and the only contact on the site is minemirrorai@gmail.com.

Three findings matter most:

- **Its angle is distinctive.** I found no Indian company that sells a physics-informed digital twin built specifically for mining safety, although several Indian firms build digital twins for other industries.
- **It is at concept stage.** The site shows a simulated control-room demo and a 3D viewer. It lists no customers, pilots, registered address or published results, and its impact numbers are placeholder counters.
- **The need in India is real.** Coal India is training staff in digital twins and AI, and India still records dozens of coal-mine fatalities every year.

2. What MineMirror AI offers

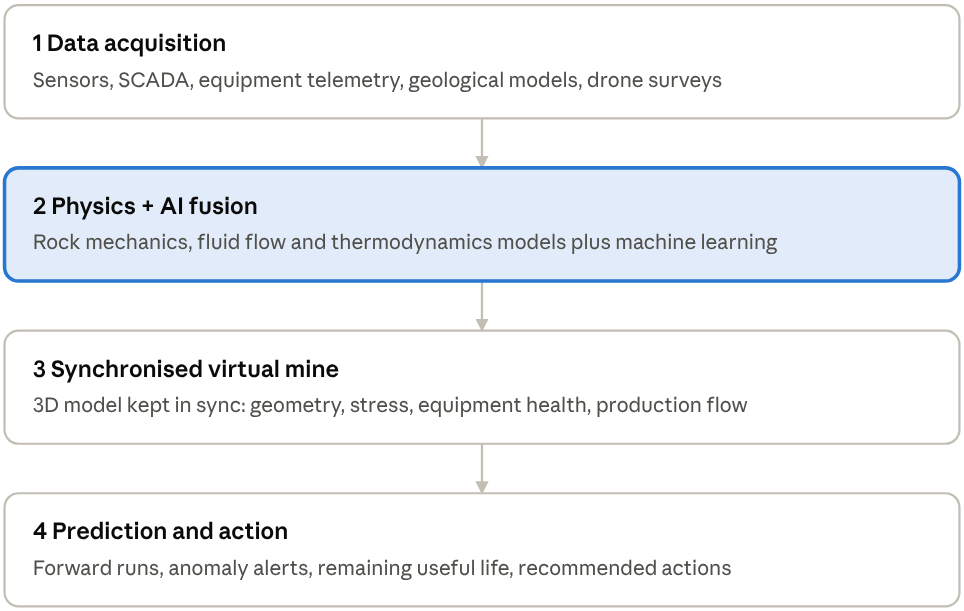
Module	What the website says it does
Predictive Safety	Early detection of geotechnical instability with lead time to act
Asset Reliability	Continuous health monitoring of dragline, haul trucks and conveyors; remaining useful life estimates
Workforce Protection	Real-time tracking of dust, vibration and air velocity for compliance and health

Module	What the website says it does
Scenario engine	"What if" runs for blast design, ventilation and haul load
3D viewer	Interactive Three.js open-pit schematic with sensor overlay

How the platform is meant to work (four steps):

1. **Data acquisition:** sensors, SCADA, equipment telemetry, geological models and drone surveys feed the twin.
2. **Physics + AI fusion:** rock mechanics, fluid flow and thermodynamics models are combined with machine learning.
3. **Synchronised virtual mine:** a 3D model stays locked to real geometry, stress, equipment health and production flow.
4. **Prediction and action:** forward simulations, anomaly detection, remaining useful life and recommended actions for operators.

The physics plus AI step lets the twin work with sparse failure data



MineMirror platform flow, 4 steps, from the MineMirror AI website

The highlighted step is the site's answer to having few recorded failures to learn from: physical laws fill the gaps that data alone cannot.

3. What makes it unique

1. Physics plus AI for sparse failure data. Mines rarely have many examples of rock failures or major breakdowns, so pure machine learning struggles. Embedding physical laws lets a model estimate conditions even with little data. This approach is already being researched at Arizona State University for a copper mine, and at IIT (ISM) Dhanbad for motor maintenance.

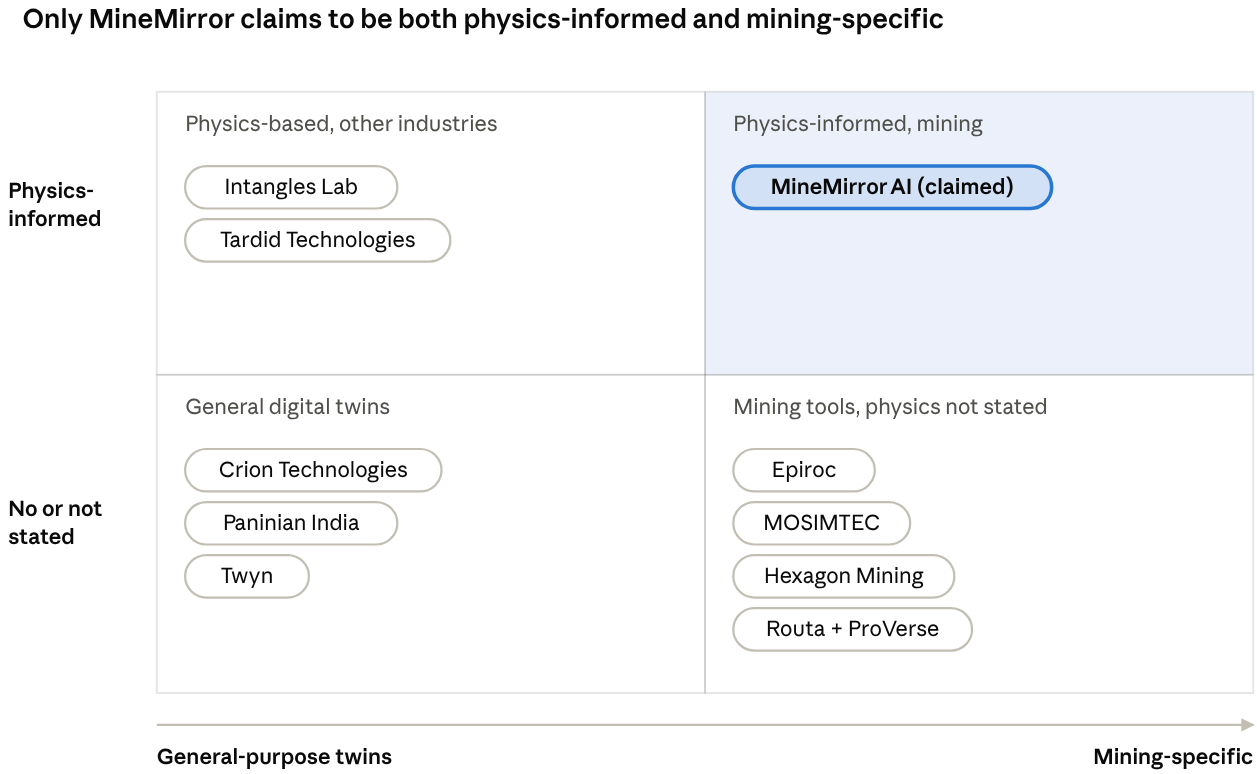
2. One platform for three risks. Most vendors cover one area: fleet (Epiroc, Caterpillar, Komatsu), planning (Hexagon, Maptek), operations simulation (MOSIMTEC), or XR training (ProVerse). MineMirror proposes ground stability, equipment health and worker exposure in a single twin.

3. Aimed at a gap in India. The closest Indian mining twin project found, Routa Digital India and ProVerse at Hindustan Copper, focuses on XR training and situational awareness. It is not described as physics-based.

4. Open collaboration model. The site invites engineers, researchers, students and mine operators to co-develop and pilot, which suits a research-led startup.

Comparison with other players

Player	Country	Focus	Physics-informed	Mining safety twin
MineMirror AI	India (claimed)	Safety, reliability, exposure	Yes (claimed)	Yes (concept)
Routa Digital India + ProVerse	India / Finland	XR training, monitoring	Not stated	Partial
Hexagon Mining	Sweden / global	Life-of-mine software	Not stated	Partial
Epiroc	Sweden / global	Collision avoidance, fleet	No	No
MOSIMTEC	USA	Process simulation (MineTwin)	No	No
Intangles Lab	India	Vehicle digital twin	Yes	No (vehicles)
Tardid Technologies	India	Physics-based AI twin	Yes	No (not mining)



positioning map, 10 companies, placed by what each one's own materials state

Placement follows what each company's own materials state, not a measured ranking. MineMirror's position is its own claim and is not yet proven.

4. What MineMirror is doing in India

Based on its website, MineMirror AI has **no documented deployments, customers or partners in India yet**. What it does publicly is:

- Present the platform concept and a simulated demo.
- Invite mining companies to pilot on a single zone or one critical asset class.
- Invite researchers and students to collaborate on geotechnical risk prediction and remaining-useful-life models.

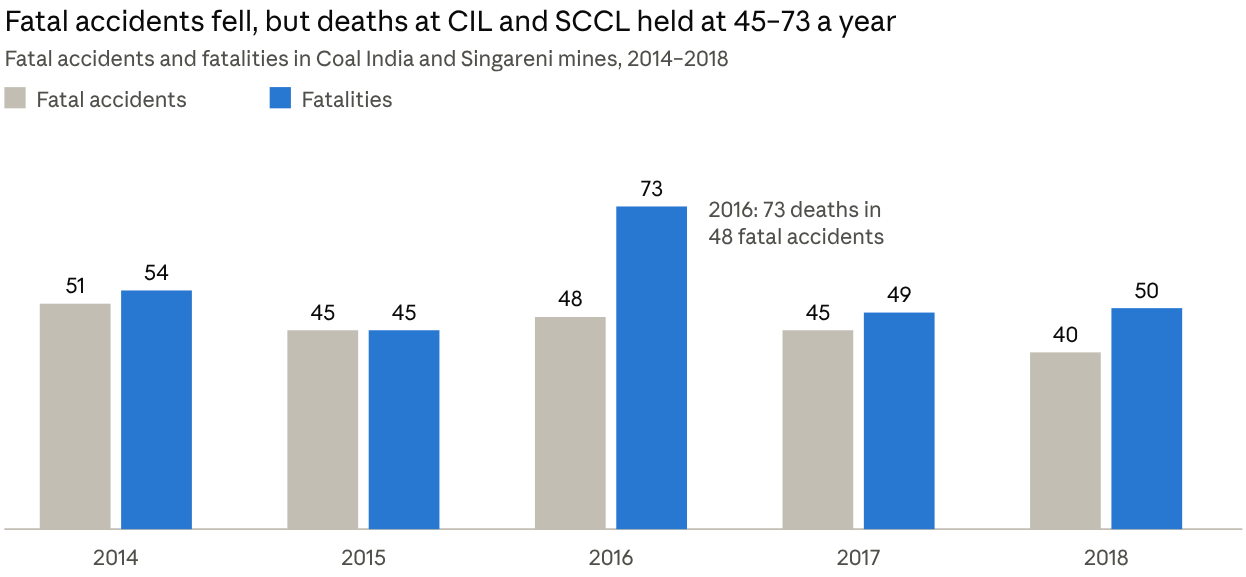
Indian landscape around it

Company / body	Location	Relevance
Routa Digital India	Noida area	Hindustan Copper mine twin proof of concept

Company / body	Location	Relevance
Hexagon Mining India	India region	Life-of-mine digital solutions
Epiroc with Hindustan Zinc	Rajasthan operations	Collision avoidance system
Coal India + IIT (ISM) Dhanbad	Dhanbad	Training on AI, IoT and digital twins
Crion Technologies	Chennai (IITM Research Park)	Industrial asset digital twins
Paninian India	Bengaluru	Aerospace digital twin
Twyn	Noida	Manufacturing digital twin

5. Why India needs it: safety data

Published figures for Coal India (CIL) and Singareni Collieries (SCCL):



Rajya Sabha answer (9 Dec 2019) on CIL and SCCL accident statistics, 2014 to 2018

More recently, the Union Mines Ministry reported **141 fatalities in coal mines between 2023 and 2025**, with Jharkhand highest at 37, followed by West Bengal. The trend is improving slowly, but deaths remain frequent, and most accident causes (roof falls, slope failure, machinery) are the very hazards a predictive twin targets.

6. Benefits

These are the **intended** benefits of the approach. MineMirror has not published validated results.

Stakeholder	Potential benefit
Mine operators	Fewer unplanned stoppages, longer equipment life, better blast and haul decisions
Workers	Earlier warning of ground and air hazards; lower exposure to dust and vibration
Regulators and DGMS	Better data for inspections and compliance evidence
Researchers and students	Real operational problems for PINN and geomechanics work
India's mining sector	Safer production growth, supporting the goal of a modern mining industry

7. Risks and gaps

- **No proof yet:** no pilot, no mine data and no published accuracy figures.
- **Unclear company status:** no registered entity, address or technical team is listed.
- **Strong competitors:** Hexagon, Epiroc and ProVerse already work inside Indian mines.
- **Integration effort:** twins depend on sensor coverage and SCADA access, which many Indian mines lack.
- **Regulatory trust:** DGMS approval and mine management confidence take time.

8. Recommended next steps

1. Register the company and publish a verifiable address and team.
2. Secure one pilot (a single zone or a single dragline or conveyor) with a mine or an institute such as IIT (ISM) Dhanbad.
3. Publish a validation case with real sensor data and measured accuracy.
4. Align the product with DGMS reporting needs.
5. Seek grants and incubation (for example Startup India or an IIT incubator).

9. Sources

- MineMirror AI website (nitishkumarroy999.github.io)
- Finnish press release on ProVerse Interactive and Routa Digital India with Hindustan Copper
- DGMS annual report 2024 and SANKET publications
- Rajya Sabha answer on CIL and SCCL accident statistics, 2014 to 2018
- ETV Bharat reports on coal mine fatalities (2023 to 2025) and Coal India training at ISM Dhanbad
- Epiroc press release on Hindustan Zinc
- Intangles funding coverage; StartUs Insights on Tardid Technologies
- Arizona State University news on the Resolution Copper digital twin